

# BIOS Bioprocess Model

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## 1 System and model description

This model aims to describe a growth system of *Pseudomonas putida* under different set-ups. The model not only investigates the microorganism's growth dynamics but also the consumption of a carbon source (substrate) and other nutrients (chemical elements). Additionally, it examines the dynamics of gases that may be involved in the process, considering both the gas phase and the dissolved phase in the liquid medium.

The model currently includes 22 equations, which are described in Section 2. To study growth, we use a basic Monod model, in which we assume that the microorganism's growth depends solely on the substrate:

$$\mu = \mu_{\max} \cdot \frac{\text{Subs}}{K_{\text{subs}} + \text{Subs}} \quad (1)$$

For the study of nutrient dynamics, the equations follow the general structure:

$$\frac{d(\text{Nutrient})}{dt} = \text{Feed} - \text{Sampling} + \text{Consumption/Generation} \quad (2)$$

The different terms are described below:

1. **Feed.** This term represents the amount of nutrients introduced into the system through the feed. If the system is being fed, it is calculated as the concentration of the nutrient in the feeding medium (mol/L) multiplied by the flow rate (L/min).
2. **Sampling.** This term accounts for the decrease in the amount of nutrients due to sampling events performed on the system. It is defined as the concentration of the nutrient in the medium (i.e., the amount of nutrient divided by the volume of medium in the system), multiplied by the sampling flow rate.
3. **Consumption/Generation.** This term represents how the organism consumes or produces nutrients. It consists of the inverse of the yield factor (mol of nutrient consumed or generated per gram of biomass), multiplied by the growth rate and the biomass.

## 2 Gas-phase and liquid-phase equations

Regarding the equations related to gases, we must distinguish between working in the liquid phase or in the gas phase, as their formulations differ. The gas-phase equations follow the structure:

$$\frac{dX_{\text{gas}}}{dt} = \text{InGasFeed} - \text{Transfer} - \text{OffGas} \quad (3)$$

The terms are defined as follows:

1. **InGasFeed.** This is the amount of gas entering the system through the supplied airflow. It is calculated as the concentration of the gas in the incoming air (mol/L) multiplied by the airflow rate (L/min).
2. **Transfer.** This term describes how gases in the air transfer into the culture medium. It consists of two parts: the  $K_L a$  parameter, which describes the mass transfer rate from the gas phase to the liquid phase (to be estimated), and the concentration difference between the gas saturation concentration and the concentration of the gas in the liquid.
3. **OffGas.** This term describes the gas leaving the system through the outgoing airflow. It is calculated as the concentration of the gas in the reactor's gas phase (mol/L) multiplied by the rate at which air exits the system (L/min).

The equation describing the behavior of gases in the liquid phase is:

$$\frac{dX_{\text{liquid}}}{dt} = \text{Transfer} + \text{Consumption/Generation} \quad (4)$$

The terms are defined as follows:

1. **Transfer.** This is the same transfer term as defined in the gas-phase equations.
2. **Consumption/Generation.** This is the same as the term used in the nutrient equations.

It is worth noting that in the case of  $\text{CO}_2$ , an additional term would be included, since in the aqueous phase this gas is hydrated to form  $\text{H}_2\text{CO}_3$ . However, this is not yet included in the model and will be incorporated at a later stage.

## 3 Model equations

This section presents the system equations. These equations may evolve as the model continues to be developed. For example, the substrate balance is given by:

$$\frac{d(\text{Subs})}{dt} = C_{\text{Subs},f} \cdot V_f - \left( \frac{\text{Subs}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{subs}}} \cdot \mu \cdot \text{Biomass} \quad (5)$$

$$\frac{d(\text{Biomass})}{dt} = \mu \cdot \text{Biomass} - \left( \frac{\text{Biomass}}{V_l} \right) \cdot V_s \quad (6)$$

$$\frac{d(\text{Ca})}{dt} = C_{\text{Ca},f} \cdot V_f - \left( \frac{\text{Ca}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Ca}}} \cdot \mu \cdot \text{Biomass} \quad (7)$$

$$\frac{d(\text{Cl})}{dt} = C_{\text{Cl},f} \cdot V_f - \left( \frac{\text{Cl}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Cl}}} \cdot \mu \cdot \text{Biomass} \quad (8)$$

$$\frac{d(\text{Co})}{dt} = C_{\text{Co},f} \cdot V_f - \left( \frac{\text{Co}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Co}}} \cdot \mu \cdot \text{Biomass} \quad (9)$$

$$\frac{d(\text{Cu})}{dt} = C_{\text{Cu},f} \cdot V_f - \left( \frac{\text{Cu}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Cu}}} \cdot \mu \cdot \text{Biomass} \quad (10)$$

$$\frac{d(\text{Fe})}{dt} = C_{\text{Fe},f} \cdot V_f - \left( \frac{\text{Fe}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Fe}}} \cdot \mu \cdot \text{Biomass} \quad (11)$$

$$\frac{d(\text{Mg})}{dt} = C_{\text{Mg},f} \cdot V_f - \left( \frac{\text{Mg}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Mg}}} \cdot \mu \cdot \text{Biomass} \quad (12)$$

$$\frac{d(\text{Mo})}{dt} = C_{\text{Mo},f} \cdot V_f - \left( \frac{\text{Mo}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Mo}}} \cdot \mu \cdot \text{Biomass} \quad (13)$$

$$\frac{d(\text{Na})}{dt} = C_{\text{Na},f} \cdot V_f + C_{\text{Na},b} \cdot V_b - \left( \frac{\text{Na}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Na}}} \cdot \mu \cdot \text{Biomass} \quad (14)$$

$$\frac{d(\text{Zn})}{dt} = C_{\text{Zn},f} \cdot V_f - \left( \frac{\text{Zn}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Zn}}} \cdot \mu \cdot \text{Biomass} \quad (15)$$

$$\frac{dK}{dt} = C_{K,f} \cdot V_f - \left( \frac{K}{V_l} \right) \cdot V_s - \frac{1}{Y_K} \cdot \mu \cdot \text{Biomass} \quad (16)$$

$$\frac{d(\text{Ni})}{dt} = C_{\text{Ni},f} \cdot V_f - \left( \frac{\text{Ni}}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{Ni}}} \cdot \mu \cdot \text{Biomass} \quad (17)$$

$$\frac{d(\text{NH}_4)}{dt} = C_{\text{NH}_4,f} \cdot V_f - \left( \frac{\text{NH}_4}{V_l} \right) \cdot V_s - \frac{1}{Y_{\text{NH}_4}} \cdot \mu \cdot \text{Biomass} \quad (18)$$

$$\frac{dP}{dt} = C_{P,f} \cdot V_f - \left( \frac{P}{V_l} \right) \cdot V_s - \frac{1}{Y_P} \cdot \mu \cdot \text{Biomass} \quad (19)$$

$$\frac{dS}{dt} = C_{S,f} \cdot V_f - \left( \frac{S}{V_l} \right) \cdot V_s - Y_S \cdot \mu \cdot \text{Biomass} \quad (20)$$

$$\frac{d(\text{CO}_2)}{dt} = [K_L a \cdot (\text{CO}_{2,\text{sat}} - \text{CO}_2) \cdot V_l] + \frac{1}{Y_{\text{CO}_2}} \cdot \mu \cdot \text{Biomass} \quad (21)$$

$$\frac{d(\text{CO}_2)}{dt} = \text{CO}_{2,\text{gas},\text{in}} \cdot V_{\text{Gas},\text{in}} - [K_L a \cdot (\text{CO}_{2,\text{sat}} - \text{CO}_2) \cdot V_l] - \left( \frac{\text{CO}_2}{V_g} \right) \cdot V_{\text{offGas}} \quad (22)$$

$$\frac{d(\text{O}_2\text{l})}{dt} = K_L a \cdot (\text{O}_{2,\text{sat}} - \text{O}_{2\text{l}}) \cdot V_l - \frac{1}{Y_{\text{O}_2}} \cdot \mu \cdot \text{Biomass} \quad (23)$$

$$\frac{d(\text{O}_2\text{g})}{dt} = \text{O}_{2,\text{gas},\text{in}} \cdot V_{\text{Gas},\text{in}} - [K_L a \cdot (\text{O}_{2,\text{sat}} - \text{O}_{2\text{l}}) \cdot V_l] - \left( \frac{\text{O}_{2\text{g}}}{V_g} \right) \cdot V_{\text{offGas}} \quad (24)$$

$$\frac{dH}{dt} = C_{H,f} \cdot V_f + C_{H,a} \cdot V_a - \left( \frac{H}{V_l} \right) \cdot V_s + \frac{1}{Y_H} \cdot \mu \cdot \text{Biomass} \quad (25)$$

$$\frac{dOH}{dt} = C_{OH,f} \cdot V_f + C_{OH,b} \cdot V_b - \left( \frac{OH}{V_l} \right) \cdot V_s \quad (26)$$

## 4 States of the system

The system states include biomass, the various nutrients in the medium broken down into their chemical elements, and the gaseous composition supplied to the culture (considering only oxygen and carbon dioxide). All system states are expressed in molar units, except for biomass, which is measured in grams of microorganism.

In this model, we use time in minutes; therefore, the units of the differential equations are mol/min, except for biomass, which is expressed in g/min.

## 5 Secondary Terms and Equations of the Model

### 5.1 Terms Related to Volumes and Flow Rates

### 5.2 Terms and Secondary Equations Related to Gases

#### 5.2.1 Terms related to gases

#### 5.2.2 Equations related to gases

Ideal gas law:

$$Pr \cdot V_{\text{gas}} = n \cdot R \cdot T_{\text{Kelvin}} \quad (27)$$

From this equation, we obtain the number of moles of gas present:

$$n = \frac{Pr \cdot V_{\text{gas}}}{R \cdot T_{\text{Kelvin}}} \quad (28)$$

| State            | Meaning                             | Units |
|------------------|-------------------------------------|-------|
| Subs             | Substrate                           | mol   |
| Biomass          | Biomass                             | g     |
| Ca               | Calcium                             | mol   |
| Cl               | Chlorine                            | mol   |
| Co               | Cobalt                              | mol   |
| Cu               | Copper                              | mol   |
| Fe               | Iron                                | mol   |
| Mg               | Magnesium                           | mol   |
| Mo               | Molybdenum                          | mol   |
| Na               | Sodium                              | mol   |
| Zn               | Zinc                                | mol   |
| K                | Potassium                           | mol   |
| Ni               | Nickel                              | mol   |
| NH <sub>4</sub>  | Ammonia                             | mol   |
| P                | Phosphorus                          | mol   |
| S                | Sulfate                             | mol   |
| CO <sub>2l</sub> | CO <sub>2</sub> in the liquid phase | mol   |
| CO <sub>2g</sub> | CO <sub>2</sub> in the gas phase    | mol   |
| O <sub>2l</sub>  | O <sub>2</sub> in the liquid phase  | mol   |
| O <sub>2g</sub>  | O <sub>2</sub> in the gas phase     | mol   |
| H                | Protons                             | mol   |
| OH               | Hydroxide                           | mol   |

Table 1: States of the model/system.

| Term                | Meaning                                        | Units |
|---------------------|------------------------------------------------|-------|
| $V_l$               | Liquid volume (volume of the liquid phase)     | L     |
| $V_g$               | Gas volume (volume of the gas phase)           | L     |
| $V_f$               | Media feed flow rate                           | L/min |
| $V_s$               | Sampling flow rate                             | L/min |
| $V_{\text{GasIn}}$  | Gas feed flow rate                             | L/min |
| $V_{\text{offGas}}$ | Volumetric flow rate of gas leaving the system | L/min |
| $V_a$               | Acid flow rate                                 | L/min |
| $V_b$               | Base flow rate                                 | L/min |

Table 2: Terms related to volumes and flow rates.

Now, we calculate the partial pressures of the gases:

$$PartialPressure_{O_2} = Fraction_{O_2} \cdot Pr \quad (29)$$

$$PartialPressure_{CO_2} = Fraction_{CO_2} \cdot Pr \quad (30)$$

Next, we calculate the saturation concentrations of the gases:

| Term                     | Meaning                                       | Value/Calculation    | Units         |
|--------------------------|-----------------------------------------------|----------------------|---------------|
| $R$                      | Gas constant (called ‘‘Rydberg’s’’ in source) | 8.31451              | kJ/mol/K      |
| $T_{sc}$                 | Temperature scale constant                    | 273.15               | K             |
| $T_{\text{Celsius}}$     | Temperature in Celsius                        | Experiment dependent | °C            |
| $T_{\text{Kelvin}}$      | Temperature in Kelvin                         | Calculated           | K             |
| $Pr$                     | Pressure                                      | 101325               | Pa            |
| $Fraction_{O_2}$         | Fraction of $O_2$ in air                      | 0.21                 | Dimensionless |
| $Fraction_{CO_2}$        | Fraction of $CO_2$ in air                     | 0.000407             | Dimensionless |
| $H_{CO_2}$               | Henry’s constant for $CO_2$                   | $3.4 \times 10^{-7}$ | mol/L·Pa      |
| $H_{O_2}$                | Henry’s constant for $O_2$                    | $1.3 \times 10^{-8}$ | mol/L·Pa      |
| $PartialPressure_{O_2}$  | Partial pressure of $O_2$                     | Calculated           | Pa            |
| $PartialPressure_{CO_2}$ | Partial pressure of $CO_2$                    | Calculated           | Pa            |
| $CSat_{CO_2}$            | Saturation concentration of $CO_2$            | Calculated           | mol/L         |
| $CSat_{O_2}$             | Saturation concentration of $O_2$             | Calculated           | mol/L         |
| $n$                      | Number of gas moles $O_2$                     | Calculated           | mol           |
| $V_{gas}$                | Gas volume (volume of gas phase)              | Experiment dependent | L             |

Table 3: Terms related to gases.

$$CSat_{O_2} = PartialPressure_{O_2} \cdot H_{O_2} \quad (31)$$

$$CSat_{CO_2} = PartialPressure_{CO_2} \cdot H_{CO_2} \quad (32)$$

It is important to note that the initial condition for the gases in the liquid phase will be equal to their saturation concentration. However, since we are working with moles and not mol/L, we must multiply by the liquid phase volume ( $V_l$ ):

$$IC_{CO_2}^l = CSat_{CO_2} \cdot V_l \quad (33)$$

$$IC_{O_2}^l = CSat_{O_2} \cdot V_l \quad (34)$$

On the other hand, if we want to know the initial condition for gases in the gas phase:

$$IC_{O_2}^g = Fraction_{O_2} \cdot n \quad (35)$$

$$IC_{CO_2}^g = Fraction_{CO_2} \cdot n \quad (36)$$

## 6 Parameters to estimate

These are the model parameters that must be estimated:

| <b>Parameter</b>  | <b>Meaning</b>            | <b>Unit</b> |
|-------------------|---------------------------|-------------|
| $\mu_{\max}$      | Maximum growth rate       | 1/min       |
| $K_{\text{subs}}$ | Substrate uptake constant | g/L         |
| $K_L a$           | Gas transfer coefficient  | 1/min       |

Table 4: Parameters to estimate.